

CHISHOLM CATHOLIC COLLEGE
Year 11 Mathematics Methods
Test 1 2024

Total Mark

/41

Section 1: Calculator - free

Name:	Time: 25 minutes maximum
Teacher:	Marks available: 21

Instructions:

- Answer all questions in the spaces provided.
- Calculators and notes are not permitted.
- Formula sheet provided.
- Where questions or parts of questions are worth more than 2 marks, justification is required.

Question 1 [1, 1, 2 = 4 marks]

a) Convert the following to radians. Express your answers in exact simplified form.

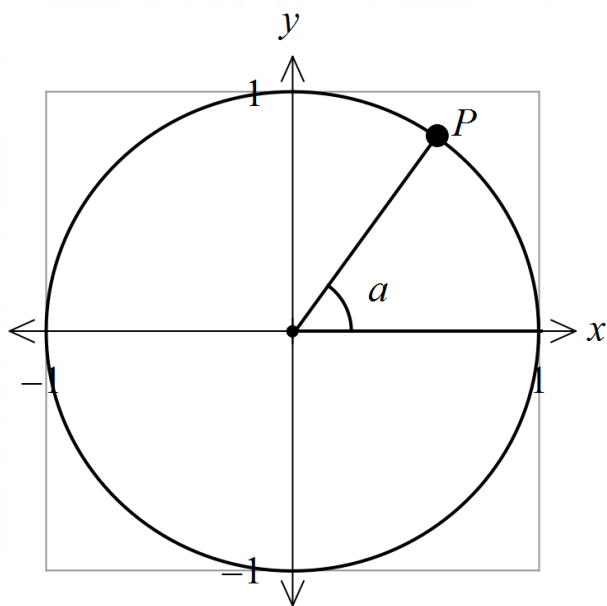
i) 120°

ii) 40°

b) Convert $\frac{5\pi}{12}$ to degrees.

Question 2 [1, 1, 1 = 3 marks]

The graph below shows an angle a in unit circle which intersects at $P(\frac{3}{5}, \frac{4}{5})$.



Determine the value of each of the following:

a) $\cos(a)$

b) $\sin(-a)$

c) $\cos(180 - a)$

Question 3 [2, 2, 2, 2 = 8 marks]

a) Determine the exact value of the following.

i) $\sin (135^\circ)$

ii) $\tan (5\pi)$

iii) $\cos \left(-\frac{5\pi}{6}\right)$

b) Determine the angle of inclination for the line given by the equation $y = \frac{\sqrt{3}}{3}x + 4$.

Question 4 [4 marks]

A bird travels 10km east of point A, then another 10km on a bearing of 030° to point B. Determine the exact distance between point A and point B.

Question 5 [2 marks]

Determine the obtuse angle in radians that will satisfy the equation $\sin x = 0.5$.



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Test 1 2024

Section 2: Calculator - assumed

Name:	Time: 25 minutes minimum
Teacher:	Marks available: 20

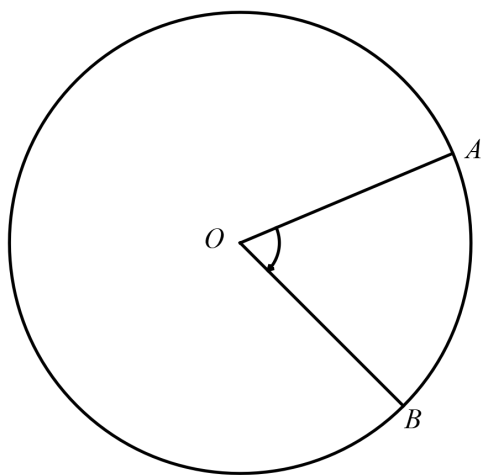
Instructions:

- Answer all questions in the spaces provided.
- Up to 3 calculators are permitted.
- 2 double-sided A4 pages of notes are permitted.
- Formula sheet provided.
- Where questions or parts of questions are worth more than 2 marks, justification is required.
- Diagrams are not drawn to scale.

Question 6 [2 marks]

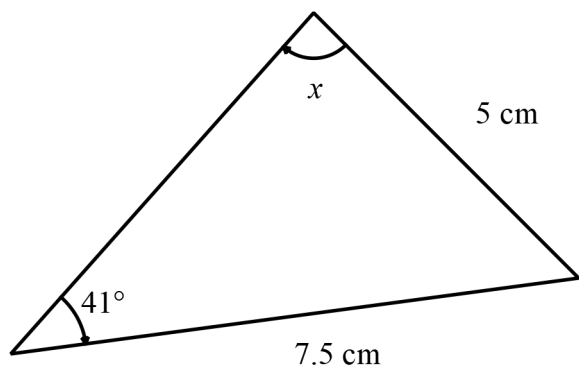
Consider the circle below where the radius is 5 m and $\angle AOB$ is $\frac{5\pi}{12}$.

Determine the length of the chord AB.



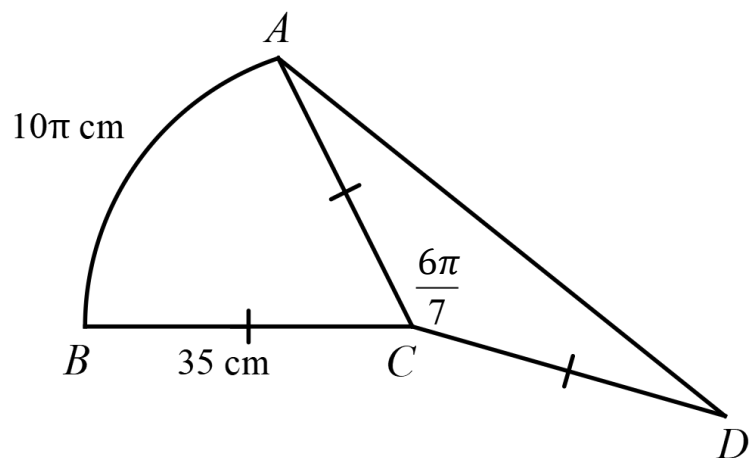
Question 7 [3 marks]

In the triangle below, determine the possible values of x to the nearest degree.



Question 8 [1, 3 = 4 marks]

Consider the following composite shape.

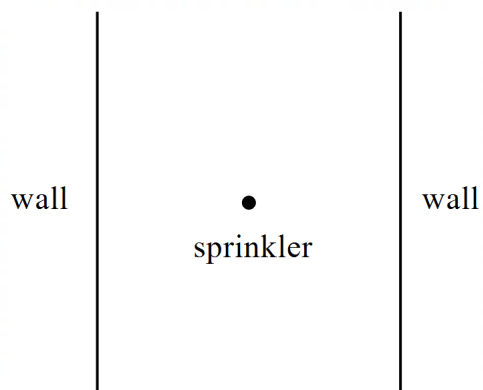


a) Determine the exact angle $\angle ACB$ in radians.

b) Calculate the area of figure $ABCD$.

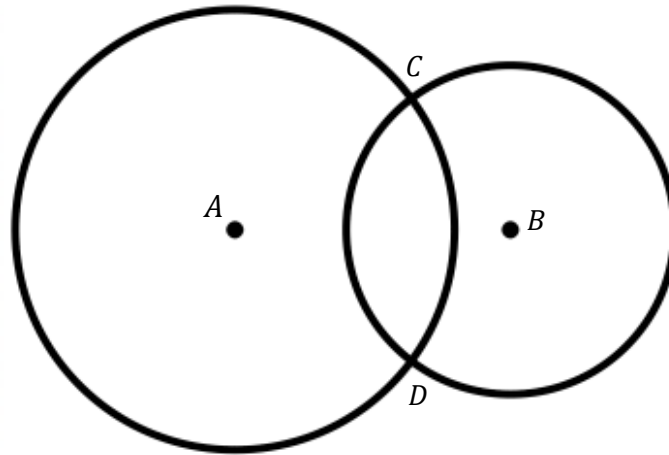
Question 9 [5 marks]

A sprinkler can propel water anywhere within a 4 m radius of its location. Given that the sprinkler is located on grass, halfway between two parallel walls 7 m apart (as shown in the diagram below), determine the area of grass watered by the sprinkler.



Question 10 [2, 4 = 6 marks]

A circle with a centre point A has a radius of 10 cm and a smaller circle with a centre point B has a radius of 7.5 cm. The circles intersect at points C and D such that the distance AB is 12.5 cm.



- a) Given $\angle ACB = \angle BDA = \frac{\pi}{2}$ radians, show that $\angle CAD = 1.29$, to two decimal places.
- b) Determine the overlapping area to 2 decimal places.